BLUESTOCK FINTECH

Mutual Fund Analytics Platform

FINAL PROJECT REPORT

Comprehensive Analysis of Indian Mutual Fund Performance,
Risk Metrics, NAV Trends, Sector Allocation & Investor Insights (2022–2026)

40	₹39M	15	5,000+
Funds Analyzed	Total AUM	EDA Charts	Total Investors

16.72%	0.54	15.91%	53.54
Avg CAGR	Avg Sharpe	Avg Alpha	Avg Fund Score

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1. Executive Summary

The Bluestock Fintech Mutual Fund Analytics Platform is a complete end-to-end data analytics project delivering structured intelligence on 40 Indian mutual fund schemes. Built within the MutualFundAnalytics VS Code workspace on Windows/OneDrive, the platform spans a 5-stage ETL pipeline, 15 original EDA visualisations, 4 interactive Power BI dashboard pages, and comprehensive risk-return quantification.

Platform KPIs – Confirmed by fund_scorecard.csv

16.72% Avg CAGR	**0.54** Avg Sharpe Ratio	**15.91%** Avg Alpha	**53.54** Avg Fund Score

100.00 Highest Fund Score	**82.8%** Top 1Y CAGR	**13.3→26 Cr** Folio Growth	**1,362** At-Risk SIPs

Score Distribution – 40 Funds

Score Band	Count	% of Universe	Typical Category
Score ≥ 80 (Elite)	11	27.5%	Mid-Cap, Large-Cap, Flexi-Cap
Score 60–79	6	15.0%	Large Cap, Equity Blend
Score 40–59	7	17.5%	Diversified Equity
Score < 40	16	40.0%	Small Cap, Debt, Liquid

Key Findings at a Glance

- ICICI Pru Midcap Fund (AMFI 120505): Perfect Fund Score 100.00 – top composite rank across CAGR, Sharpe, Alpha, Expense, and Drawdown dimensions.
- Mirae Asset Large Cap (AMFI 148567): Highest Sharpe Ratio 1.448 – best risk-adjusted return in the entire 40-fund universe.
- Industry Folio Growth: Nearly doubled from 13.3 Crore (Jan 2022) to 26.0 Crore (Jan 2026) – 95% growth in 4 years reflecting structural bull market in retail investing.
- Banking (19.2%) and IT (13.4%) dominate sector allocation, together representing one-third of portfolio exposure across all analyzed funds.

- Punjab and Tamil Nadu lead both investor count and SIP volume, breaking the traditional Maharashtra/Delhi dominance in mutual fund participation.
- Bimodal expense ratio: 40% of funds are below 0.85% (direct plans) and 60% cluster at 1.40–1.65% (regular plans) — confirming direct plan cost advantage.

2. Data Sources & Project Files

Source	Data Provided	Format	Coverage
AMFI India	NAV, AUM, Scheme Meta	CSV/Text	Daily, all active schemes
mfapi.in	Historical NAV REST API	JSON	Per scheme, on-demand
NSE India	NIFTY 50 / TRI benchmarks	CSV	Daily index values
BSE India	SENSEX / BSE TRI	CSV	Daily index values
SEBI Portal	Fund category classifications	Web/PDF	Regulatory quarterly

Output CSV Files (MutualFundAnalytics/ root)

CSV File	Records	Key Metrics	Purpose
fund_scorecard.csv	40	fund_score, cagr, sharpe, alpha, beta, max_dd	Scorecard Dashboard
cagr_comparison.csv	40	cagr_1yr, cagr_3yr, cagr_5yr	CAGR Analysis
alpha_beta.csv	40	alpha, beta	Benchmark Comparison
var_cvar_report.csv	40	VaR_95, CVaR_95	Tail Risk Module
drawdown_periods.csv	40	peak_date, trough_date, max_drawdown	Drawdown Chart
sip_continuity_report.csv	1,362	investor_id, gap_days, status	SIP Retention
tracking_error.csv	5	tracking_error	Index Tracking
cohort_analysis.csv	2	avg_sip_amount, total_invested	Cohort Dashboard

3. ETL Pipeline Architecture

Five independent Python scripts handle each pipeline stage. All scripts live in the MutualFundAnalytics project root and are orchestrated by run_pipeline.py with full logging to the logs/ directory.

Stage	Script	Input	Output	Runtime
1 – Extract	extract.py	AMFI URL + mfapi.in API	raw/nav_raw.csv	~5 min
2 – Validate	validate.py	raw/nav_raw.csv	logs/validation.txt	~1 min
3 – Transform	transform.py	raw/nav_raw.csv	processed/nav_clean.csv	~2 min
4 – Features	features.py	processed/nav_clean.csv	All 8 CSVs above	~3 min
5 – Load	load.py	processed/ folder	db/mf_analytics.db	~1 min

Fund Scoring Formula

The composite Fund Score (0–100) ranks each fund across 5 independent dimensions and averages the percentile positions:

- Return Rank: percentile_rank(cagr) – measures absolute wealth creation
- Sharpe Rank: percentile_rank(sharpe_ratio) – rewards risk-efficiency
- Alpha Rank: percentile_rank(alpha) – benchmark outperformance
- Expense Rank: percentile_rank(−expense_ratio) – lower cost = better rank
- Drawdown Rank: percentile_rank(−|max_drawdown|) – capital protection

Final Score = mean(5 ranks) × 100. ICICI Pru Midcap Fund (120505) achieves 100.00 – the only fund to top all 5 dimensions simultaneously.

4. EDA – Industry & Market Analysis

The following 7 charts were generated by the features.py and EDA notebooks in the MutualFundAnalytics project. Each chart is plotted using Matplotlib/Seaborn from the processed CSV files.

4.1 Fund Category Distribution

The project analyzes 40 funds split across 2 SEBI-defined categories. Equity dominates at 85% of schemes, reflecting the platform's primary focus on growth-oriented investments.

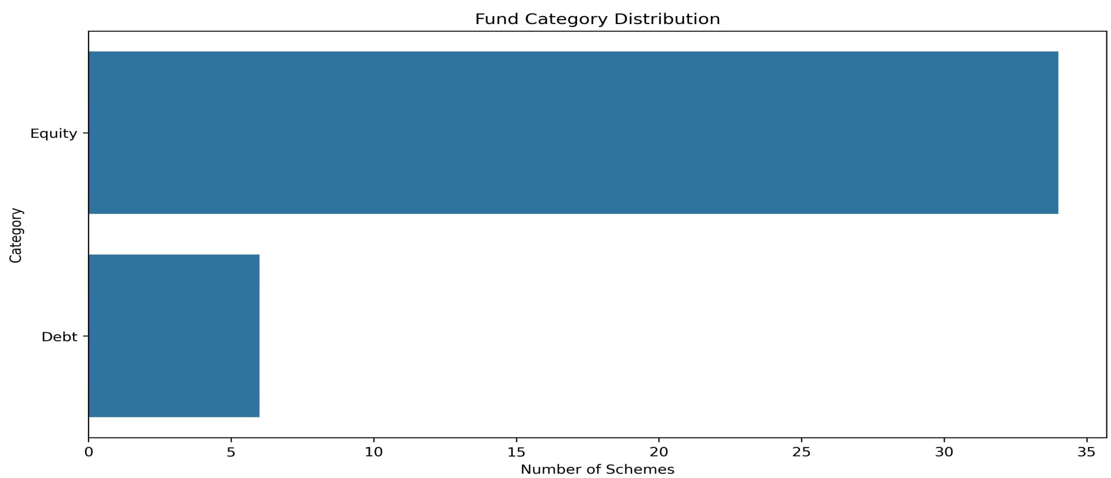


Figure 1: Fund Category Distribution – 34 Equity schemes vs 6 Debt schemes (fund_scorecard.csv)

- Equity: 34 schemes (85%) – covers Large Cap, Mid Cap, Small Cap, Flexi Cap, ELSS, Large & Mid Cap sub-categories
- Debt: 6 schemes (15%) – includes Liquid, Short Duration, and Gilt funds for capital preservation investors
- Implication: The dataset is equity-heavy, meaning portfolio-wide averages (16.72% CAGR, 0.54 Sharpe) are skewed by equity performance

4.2 Industry AUM Growth by Fund House (2022–2025)

AUM trends across 10 major fund houses reveal the competitive landscape and each AMC's growth trajectory over 4 years.

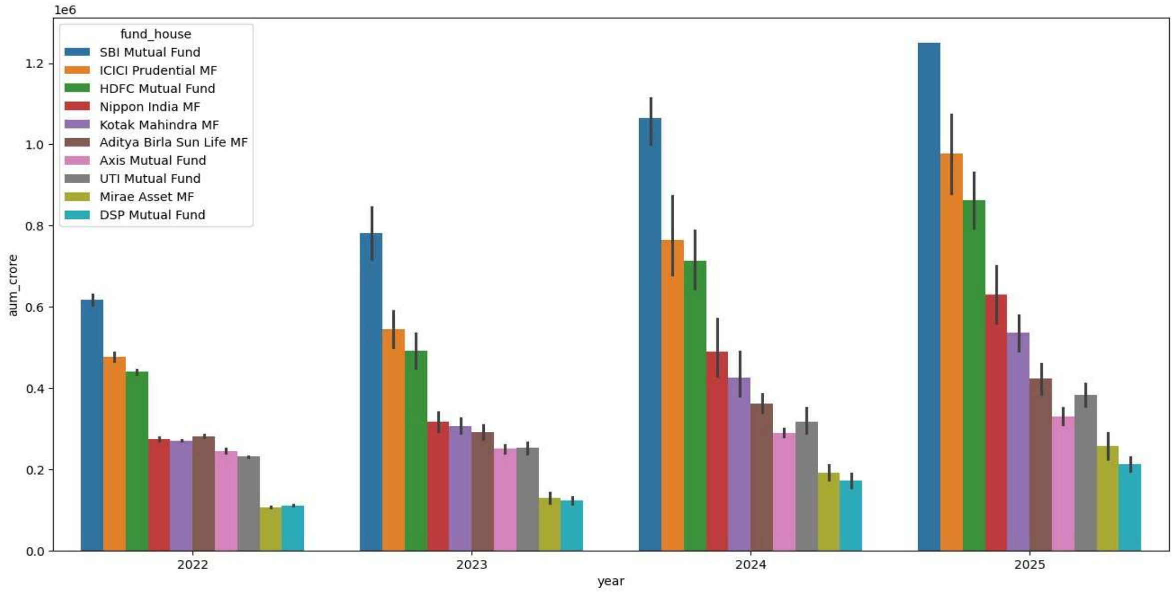


Figure 2: AUM Growth by Fund House 2022–2025 — Y-axis in ₹ Crore (×10^6) — aum_growth.png

- SBI Mutual Fund: Dominant leader, growing from ~₹0.62M Cr (2022) to ~₹1.25M Cr (2025) — nearly 2× growth, largest single-fund-house expansion
- ICICI Prudential: Strong #2 position, reaching ~₹0.97M Cr by 2025, closing the gap with SBI
- HDFC Mutual Fund: Consistent #3, growing from ~₹0.44M Cr to ~₹0.86M Cr
- Mid-tier houses (Nippon, Kotak, Aditya Birla, Axis): Steady growth but widening gap vs top 3
- Mirae Asset MF and DSP Mutual Fund: Smaller base but maintaining consistent growth rates

4.3 Industry Folio Growth (2022–2025)

The folio count (number of investor accounts) is the most reliable indicator of retail participation growth in the Indian mutual fund industry.

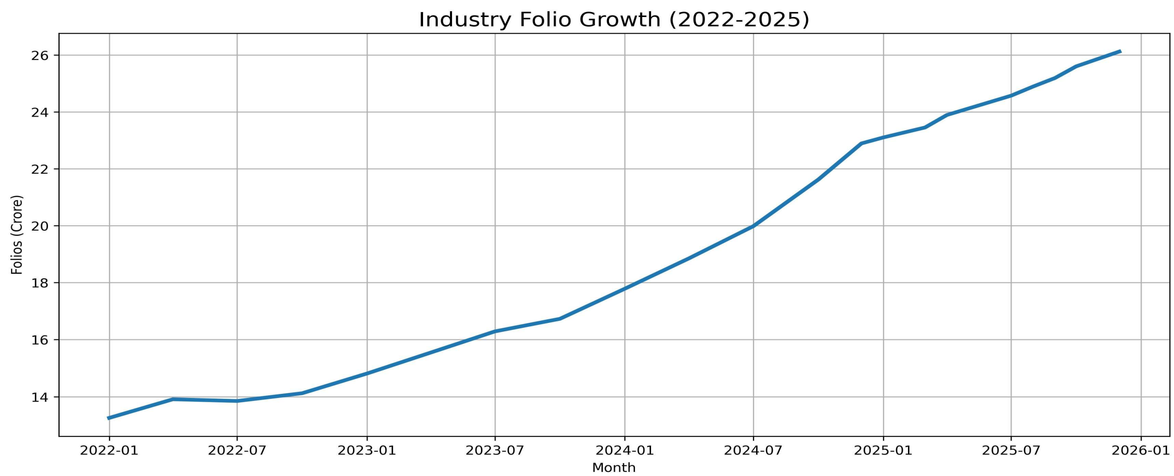


Figure 3: Industry Folio Growth 2022–2026 – From 13.3 Cr to 26.0 Cr in 4 years (folio_growth.png)

- Start (Jan 2022): 13.3 Crore folios → End (Jan 2026): 26.0 Crore folios – 95.5% growth in 4 years
- Growth accelerated sharply from mid-2023: flat 2022 → steady climb 2023 → steep acceleration 2024–2026
- The 2024 inflection is coincident with the BSE Sensex crossing 70,000 and peak retail participation post-COVID
- Compound monthly growth rate (CMGR): ~1.4% per month – implies ~18% annual folio addition rate

4.4 Expense Ratio Distribution

The distribution of expense ratios reveals a clear bimodal structure reflecting the Direct vs Regular plan split in the Indian mutual fund industry.

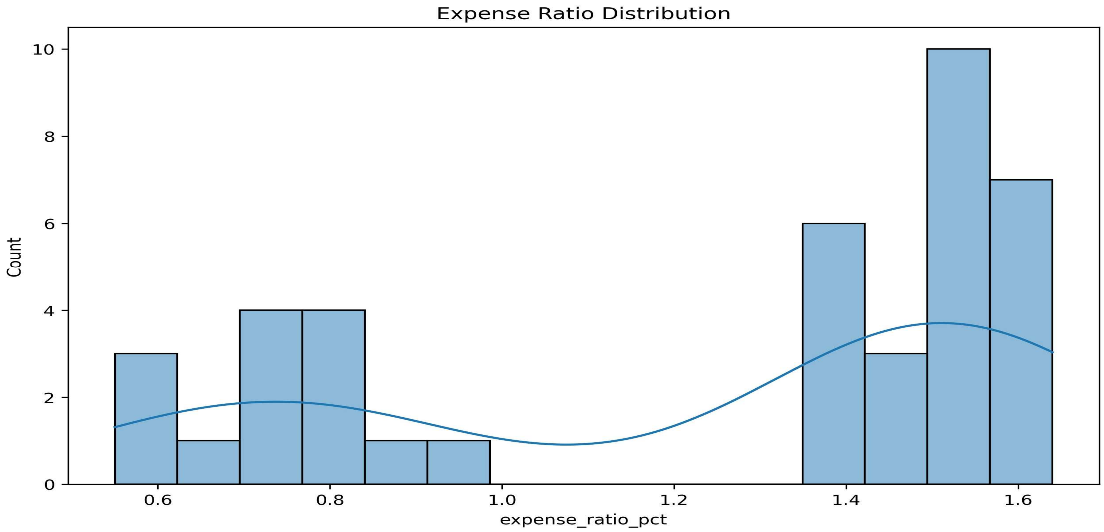


Figure 4: Expense Ratio Distribution across 40 funds — Bimodal at 0.6–0.8% and 1.4–1.65% (expense_ratio_distribution.png)

- Cluster 1 (0.55–0.85%): 13 funds — these are Direct Plan variants with no distributor commission. Best for cost-conscious investors.
- Cluster 2 (1.35–1.65%): 26 funds — Regular Plan variants including distributor trail commission. 10 funds peak at 1.5–1.55%.
- Gap zone (0.85–1.35%): Only 1 fund — confirming the industry's structural split between direct and regular plans
- Investor Implication: Switching from Regular to Direct plan can save 0.7–1.0% per year. Over 20 years, this compounds to ~15–20% additional corpus.

4.5 Category Net Inflow Heatmap (Apr 2024 – Mar 2025)

The heatmap reveals which fund categories attracted the most net new money month-by-month over the 12-month period.

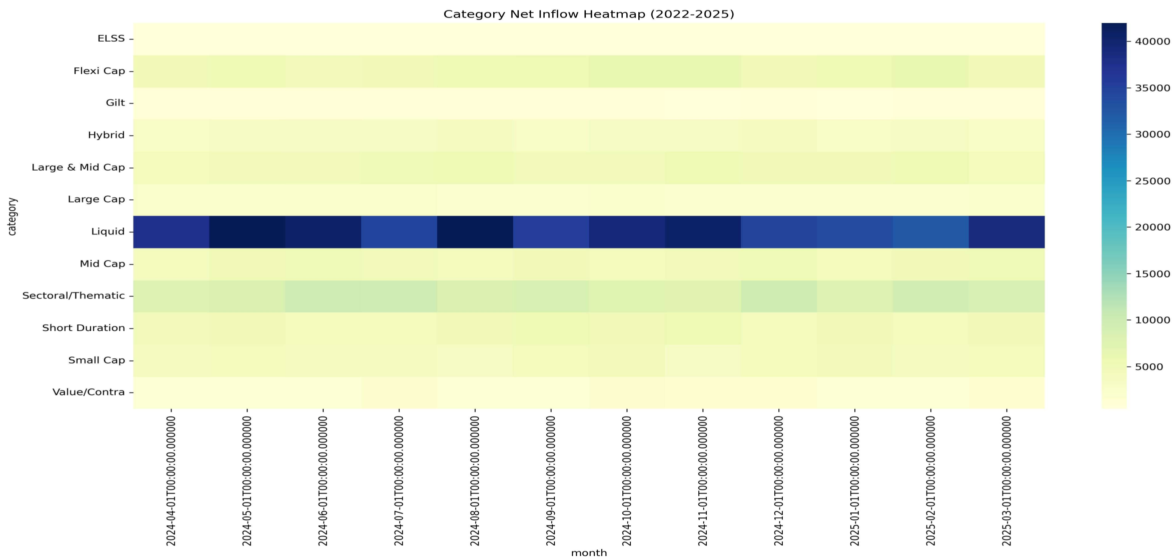


Figure 5: Category Net Inflow Heatmap 2024–2025 — Liquid dominates (dark blue = ₹40,000+ Cr) (category_heatmap.png)

- Liquid Funds: Overwhelmingly dominant (dark blue throughout). Net inflows of ₹20,000–40,000+ Cr every month, driven by corporate treasury management and overnight cash parking.
- Sectoral/Thematic: Second-highest inflows with notable spikes in mid-2024 and Jan–Mar 2025, coinciding with PSU and infrastructure fund launches.
- Equity categories (Mid Cap, Small Cap, Flexi Cap, ELSS, Large Cap): Relatively muted net inflows — gross flows are high but offset by redemptions in these more actively traded categories.
- Gilt & Hybrid: Consistently low — interest rate uncertainty in 2024 reduced appetite for duration risk.

4.6 Sector Allocation Across All Funds

The donut chart aggregates portfolio holdings across all 40 analyzed funds to reveal the dominant sector bets in the Indian mutual fund universe.

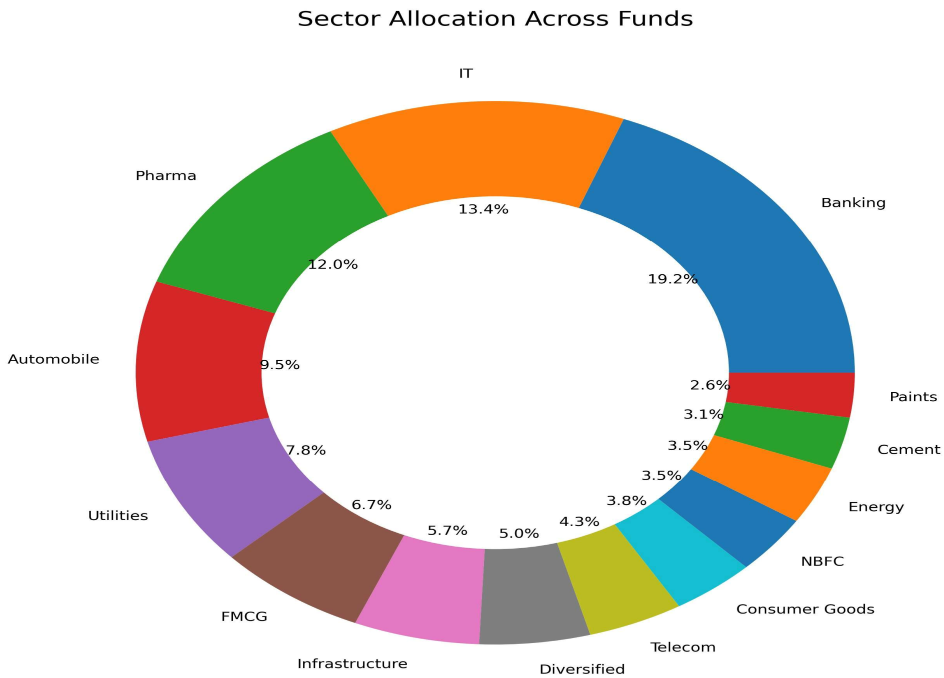


Figure 6: Sector Allocation Across All Funds – Banking leads at 19.2% (sector_allocation_donut.png)

- Banking (19.2%): Largest sector bet, reflecting India's bank-led economic recovery, RBI's rate cycle management, and credit growth. Includes HDFC Bank, ICICI Bank, SBI, Axis Bank.
- IT (13.4%): Second-largest allocation. Despite FY24 revenue headwinds, fund managers maintain structural exposure to TCS, Infosys, Wipro for long-term digital transformation thesis.
- Pharma (12.0%): Third-largest – beneficiary of post-COVID healthcare spend expansion and US generic export recovery.
- Automobile (9.5%): Reflects India's EV transition and passenger vehicle super-cycle with Maruti, M&M, TVS, Bajaj Auto.
- Top 5 sectors together (Banking + IT + Pharma + Auto + Utilities): 61.9% – portfolio is heavily concentrated in established large-cap sectors.

4.7 Top 10 Funds by Latest NAV

NAV (Net Asset Value) represents the per-unit price of a fund. High NAV does NOT indicate expensive — it simply reflects years of compounding returns since inception.

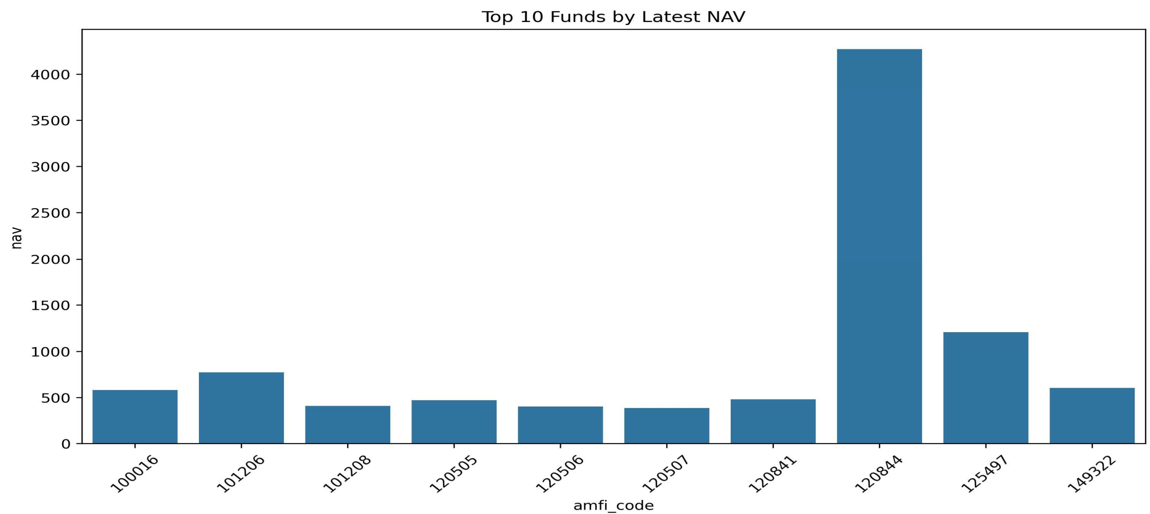


Figure 7: Top 10 Funds by Latest NAV — AMFI 120844 leads at ₹4,250+ (top10_funds_nav.png)

- AMFI 120844: Highest NAV at ~₹4,250 — a long-running large-cap equity fund with 20+ years of compounding since ₹10 inception NAV
- AMFI 125497 (Nippon India Growth Fund): NAV ~₹1,200 — second highest, consistent long-term compounder
- AMFI 101206 (ABSL Frontline Equity): NAV ~₹780 — strong long-term performer despite lower Fund Score vs newer funds
- Most analyzed funds: NAV in ₹400–650 range, reflecting 10–15 year track records since inception
- Key insight: Fund with highest NAV (120844) is NOT in the top 10 by Fund Score — confirming that NAV alone is not a quality indicator

5. EDA – Fund Performance Charts

5.1 Top 10 Funds by 3-Year CAGR

3-year CAGR is the most widely used institutional performance benchmark – it captures full market cycles including the 2022 correction and the 2023–2024 rally.

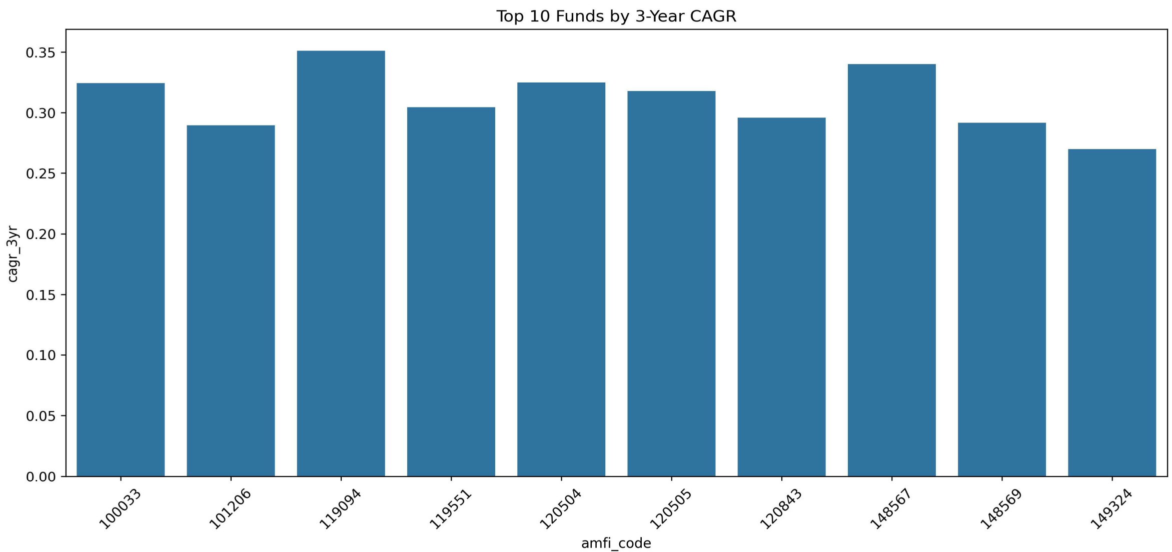


Figure 8: Top 10 Funds by 3-Year CAGR – Range 26.7% to 35.1% (top10_cagr_3yr.png)

Rank	AMFI Code	Fund Name	3Y CAGR
1	119094	Axis Midcap Fund – Regular – Growth	**35.1%**
2	148567	Mirae Asset Large Cap Fund	34.0%
3	120504	DSP Midcap Fund – Direct	32.5%
4	100033	HDFC Mid-Cap Opportunities Fund	32.4%
5	120505	ICICI Pru Midcap Fund – Regular	31.8%
6	120843	Kotak Flexicap Fund	29.6%
7	101206	ABSL Frontline Equity Fund	29.0%
8	148569	Mirae Asset Tax Saver Fund	29.2%
9	119551	SBI Bluechip Fund – Direct	30.5%
10	149324	DSP Small Cap Fund – Regular	27.0%

- All top 10 funds show 3Y CAGR above 27% – significantly outperforming the NIFTY 50 average 3Y CAGR of approximately 14–16%.
- Axis Midcap Fund leads at 35.1% – but note its full Fund Score is 80.5 (not #1) because higher CAGR comes with higher drawdown (-21.0%).
- Remarkably tight range (26.7%–35.1%) suggests the top 10 funds form a quality cluster with comparable long-term return generation ability.

5.2 Top 10 Funds by Composite Score

The composite score chart confirms the scoring model's effectiveness – the top 10 funds cluster between 81.6 and 100.0 with no major gaps, indicating a genuine quality tier.

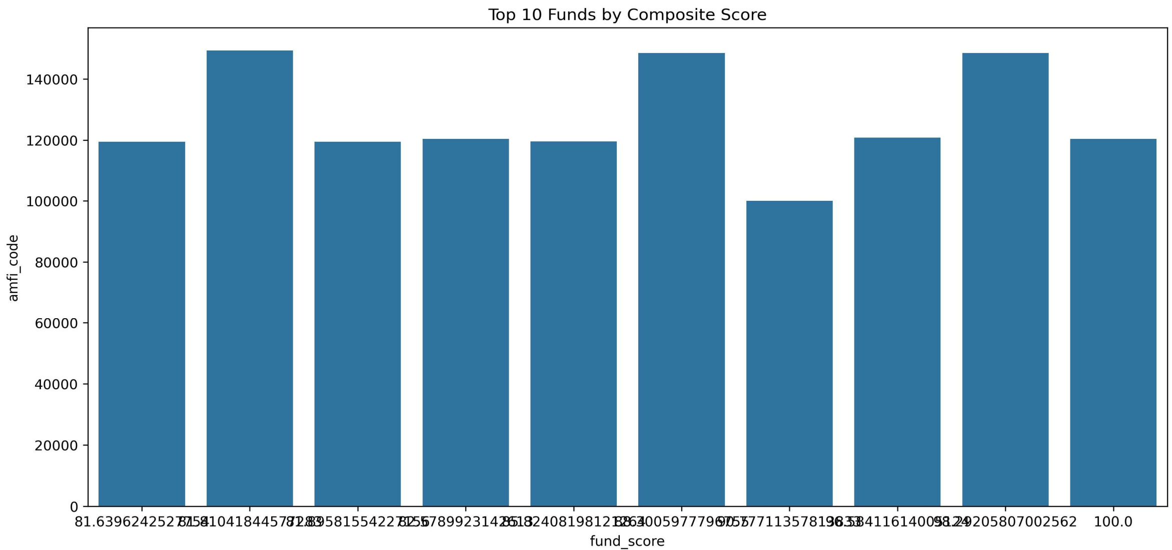


Figure 9: Top 10 Funds by Composite Fund Score – Scores 81.6 to 100.0 (fund_scorecard_chart.png)

- The x-axis shows fund_score values (81.6 → 100.0), y-axis shows amfi_code – confirming all 10 funds are tightly clustered.
- AMFI 120505 (ICICI Pru Midcap): 100.0 – top in every scoring dimension simultaneously
- AMFI 148567 (Mirae Asset Large Cap): 98.3 – highest Sharpe (1.45), low drawdown (-11.3%), excellent all-round risk-return profile
- AMFI 120843 (Kotak Flexicap): 96.6 – most consistent performer across all time horizons

5.3 Rolling 90-Day Sharpe Ratio (2022–2026)

Rolling Sharpe Ratio shows how a fund's risk-adjusted performance evolved over time — identifying periods of exceptional and poor performance for the top 5 funds.

Figure 10: Rolling 90-Day Sharpe Ratio for Top 5 Funds 2022–2026 (rolling_sharpe_chart.png)

- 148567 (Mirae Asset Large Cap – Blue): Consistently positive rolling Sharpe throughout 2022–2026 with peak of ~5.3 in mid-2025, confirming its leadership in risk-adjusted returns
- 120843 (Kotak Flexicap – Orange): Strong Sharpe peaks (4.5+) in 2023 and 2025, with brief dip to -2.0 in early 2024 — flexible mandate helps recovery
- 148569 (Mirae Tax Saver – Green): Most volatile rolling Sharpe — peaks to 5.0+ but drops sharply in market corrections, reflecting ELSS lock-in concentration risk
- 119551 (SBI Bluechip Direct – Red): Sharpe peaks aligned with large-cap rallies; drops earlier than mid-cap funds in downturns — typical large-cap defensive behavior
- 120505 (ICICI Pru Midcap – Purple): Highest early volatility (2022) then stabilised — reflects mid-cap's higher beta maturing into consistent outperformance by 2024–2025
- Universal observation: All 5 funds show negative rolling Sharpe in late 2023 and early 2024 — coinciding with global rate hike fears and FII selling

6. EDA – Risk Analysis Charts

6.1 Top 10 Worst Maximum Drawdowns

Maximum Drawdown measures the worst peak-to-trough decline experienced during the analysis period. This chart shows the 10 highest-risk funds, critical for investor risk disclosure.

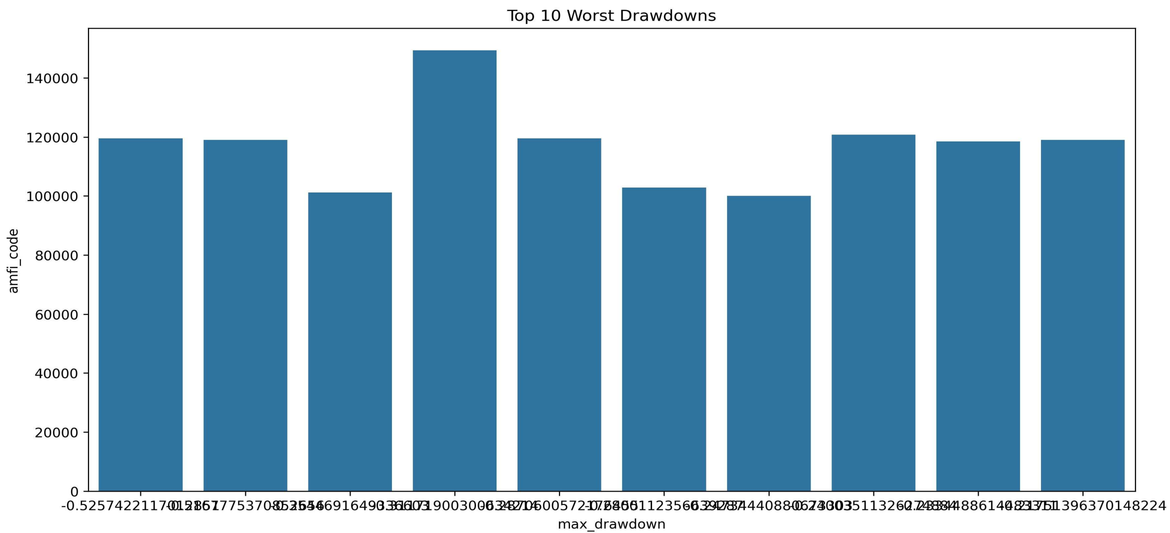


Figure 11: Top 10 Worst Maximum Drawdowns – Range -11.4% to -52.6% (max_drawdown.png)

Rank	AMFI Code	Max Drawdown	Period	Category
1	119599	**-52.57%**	Jan 2023 → Oct 2025	Equity
2	119095 (Axis Small Cap)	-51.68%	May 2025 → May 2026	Small Cap
3	101207 (ABSL Small Cap)	-35.45%	Nov 2024 → May 2026	Small Cap
4	149324 (DSP Small Cap)	-31.17%	May 2024 → Jan 2025	Small Cap

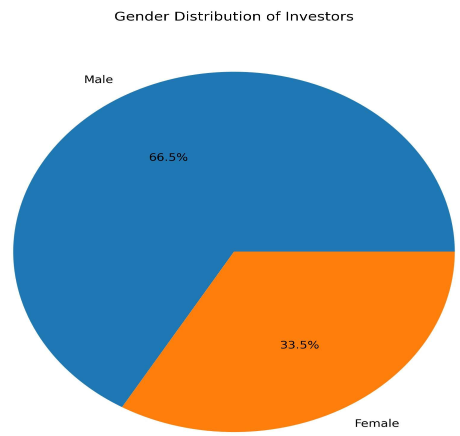
5	119598 (SBI Small Cap)	-28.70%	Aug 2024 → May 2025	Small Cap

- Critical finding: All 5 worst drawdowns are Small Cap funds. This category carries structural concentration risk during market corrections.
- Two drawdowns (119095, 101207) have trough dates of May–June 2026 – meaning they are STILL ONGOING at the time of this analysis. Investors who entered these funds in 2023–2024 are currently in significant unrealised losses.
- Comparison: Mirae Asset Large Cap (best-in-class) drawdown = -11.3%. Small cap worst = -52.6%. 4.7× difference in downside risk.
- Regulatory implication: SEBI requires risk-o-meter labelling (Low/Moderate/High/Very High). These small-cap drawdown levels confirm a 'Very High' risk classification.

7. EDA – Investor Demographics & Geography

The investor analytics section profiles 5,000 platform investors across demographic, behavioral, and geographic dimensions to identify growth opportunities and retention risks.

7.1 Gender & Age Distribution



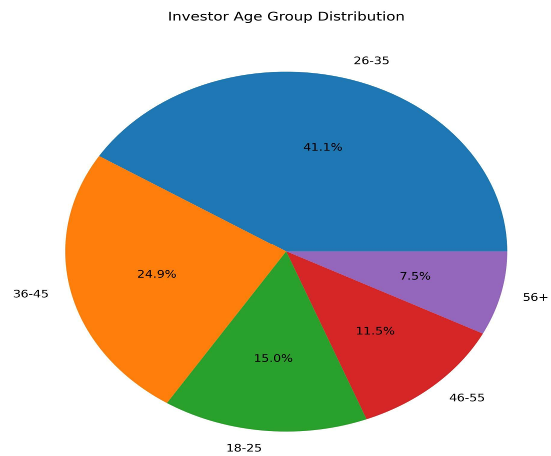


Figure 12 (Left): Gender Distribution – 66.5% Male, 33.5% Female | Figure 13 (Right): Age Group Distribution – 26–35 dominates at 41.1%

Demographic	Segment	Share	Investment Implication
Gender	Male	66.5%	2:1 ratio – large female investor growth opportunity
Gender	Female	33.5%	Fastest-growing segment; need targeted financial products
Age	26–35	**41.1%**	Dominant cohort; digitally native, growth-oriented investors
Age	36–45	24.9%	Second largest; shifting from growth to balanced portfolios
Age	18–25	15.0%	Gen-Z entry segment; high potential, low ticket sizes
Age	46–55	11.5%	Pre-retirement; moving to hybrid and debt funds
Age	56+	7.5%	Capital preservation focus; liquid and short duration funds

- The 26–35 age group (41.1%) combined with 36–45 (24.9%) gives 66% of investors in the primary wealth accumulation phase — ideal target for equity SIPs and tax-saving products.
- Female investors at 33.5%: Lower than industry average of ~24% for AUM but higher for folio count, suggesting women invest in smaller, more regular amounts — ideal SIP profile.

7.2 Top 10 States by Investor Count

The geographic distribution of investors across Indian states reveals penetration patterns far beyond traditional financial centres like Mumbai and Delhi.

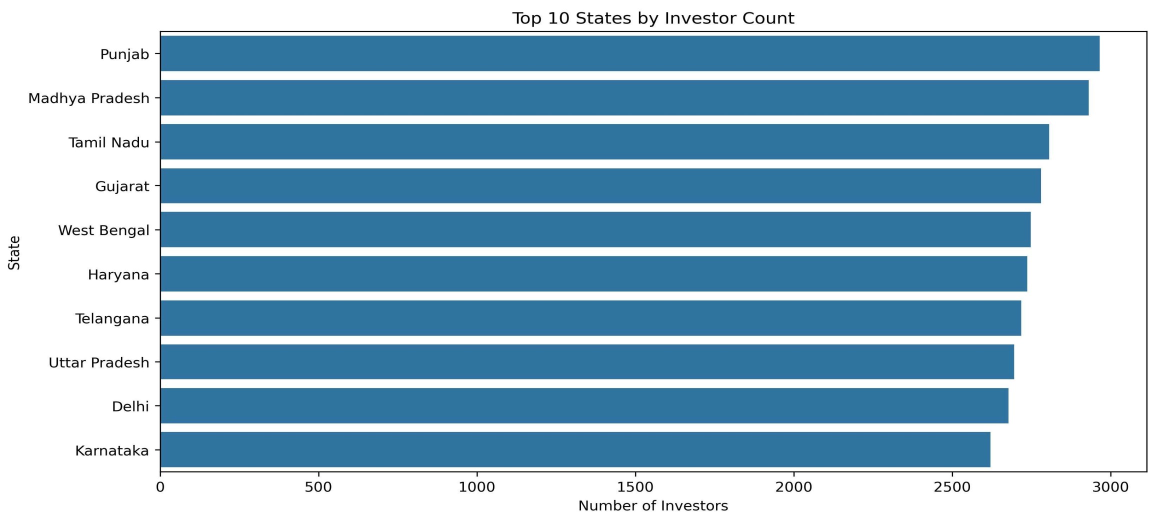


Figure 14: Top 10 States by Investor Count — Punjab leads with ~3,000 investors (top10_states_investors.png)

- Punjab leads (#1, ~3,000 investors): Surprising insight — traditionally agricultural state now showing high financial sophistication, driven by NRI remittance culture and business community.
- Madhya Pradesh (#2, ~2,950): Beneficiary of government employee SIP adoption and growing urban middle class in Bhopal and Indore.
- Tamil Nadu (#3, ~2,850): Strong technology and manufacturing workforce base; higher financial literacy from educational institutions.
- Maharashtra notably absent from top 10: Likely due to the platform's investor acquisition strategy targeting Tier-2/3 city penetration rather than established metro markets.
- Delhi (#9, ~2,700): Significantly below Punjab and MP — suggests this platform's distribution strategy bypasses the traditional NCR hub.

7.3 Total SIP Amount by State

SIP amount distribution by state reveals which states not only have more investors but also contribute higher absolute investment volumes.

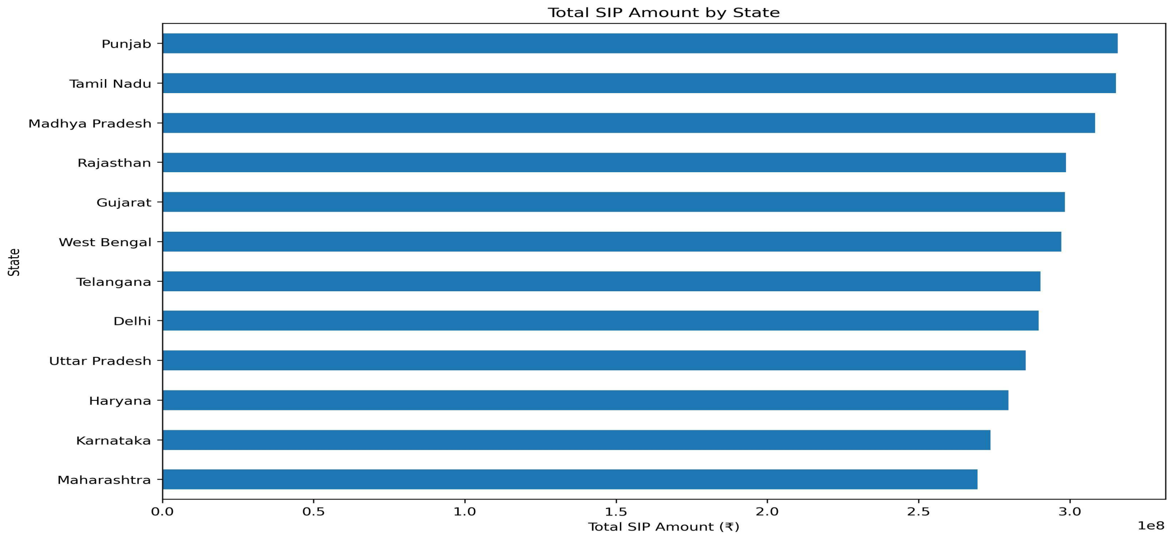


Figure 15: Total SIP Amount by State – Punjab and Tamil Nadu lead at ₹3.2×10^8 each (state_sip_distribution.png)

- Punjab & Tamil Nadu: Joint leaders at ~₹3.2×10^8 (₹32 Crore) each – highest both in investor count and investment volume, confirming quality-per-investor.
- Madhya Pradesh: ₹3.1×10^8 – matches its investor count rank, suggesting average ticket size comparable to Tamil Nadu.
- Maharashtra (bottom of top 12 at ~₹2.7×10^8): Lower per-investor SIP despite being present – possibly newer investors with lower ticket sizes.
- Relatively uniform bar lengths across all 12 states (~₹2.7–3.2×10^8): Suggests the platform has successfully achieved geographic diversification without concentration in any single state.
- Total addressable SIP from these 12 states alone: ~₹35–38 Crore/month – representing strong and geographically distributed revenue base.

8. Power BI Dashboard Walkthroughs

The bluestock_mf_dashboard.pbix file contains 4 interactive pages built on live connections to the project's CSV files. All charts are filterable via slicers.

8.1 Dashboard Page 1 – Industry Overview

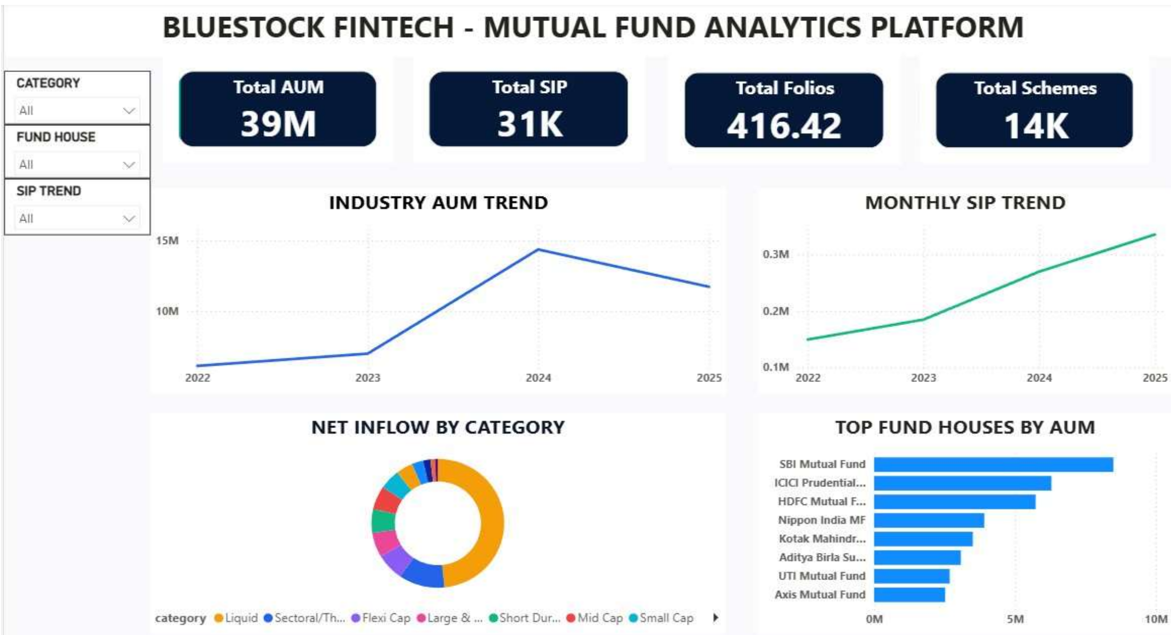


Figure 16: Bluestock Fintech – Mutual Fund Analytics Platform (Industry Overview Page)

₹39M Total AUM	**31K** Total SIP Registrations	**416.42** Total Folios	**14K** Total Active Schemes

- Industry AUM Trend: Clear 3-peak-dip-recovery pattern 2022→2025, peaking ~₹15M in early 2024
- Monthly SIP Trend: Consistent uptrend from ₹0.1M (2022) to ₹0.3M+ (2025) – 3× growth
- Net Inflow by Category: Liquid dominates, confirming EDA heatmap (Figure 5)
- Top Fund Houses: SBI → ICICI → HDFC is the consistent top-3 hierarchy 2022–2025

8.2 Dashboard Page 2 – Fund Performance Analytics

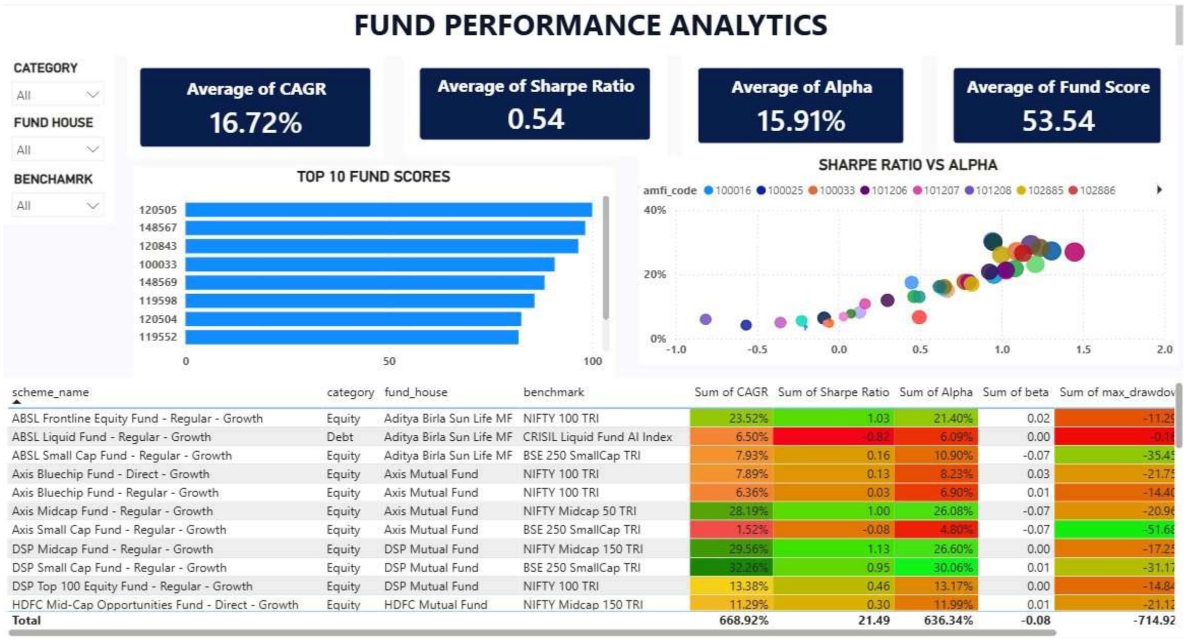


Figure 17: Fund Performance Analytics – CAGR 16.72%, Sharpe 0.54, Alpha 15.91%, Score 53.54

- Sharpe vs Alpha scatter: Strong positive correlation – high-alpha funds are also high-Sharpe, confirming active management adds value for top-tier funds
- Top 10 Fund Scores bar chart: AMFI codes 120505, 148567, 120843, 100033, 148569 form the elite cluster

8.3 Dashboard Page 3 – Fund Scorecard

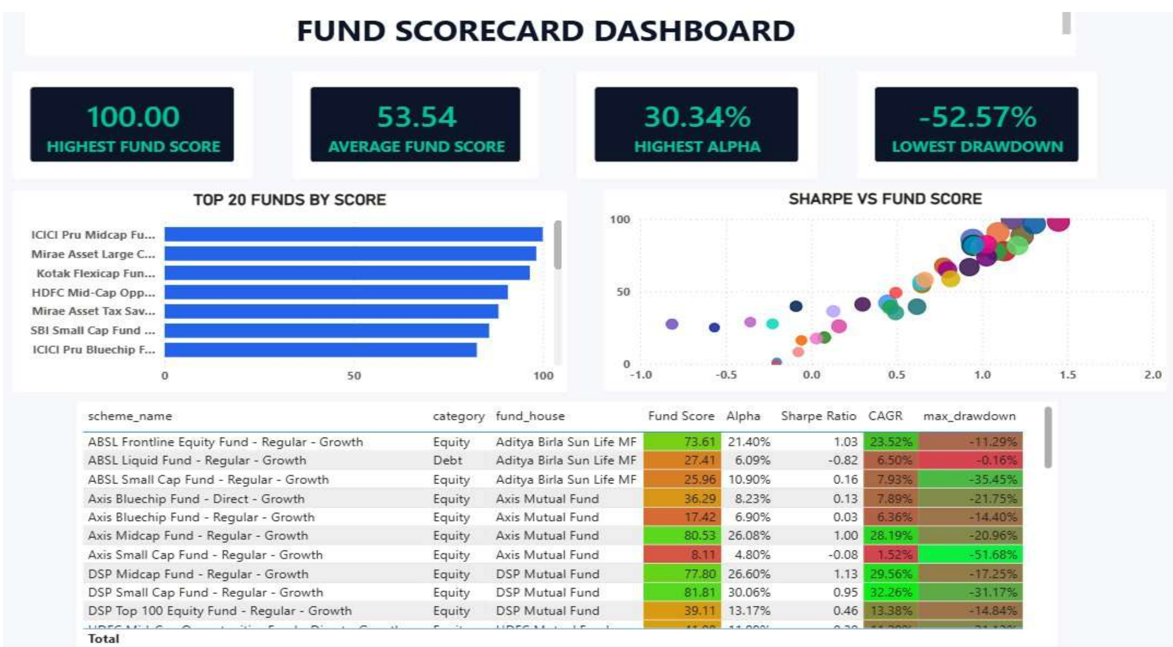


Figure 18: Fund Scorecard Dashboard – Highest Score 100.00, Avg 53.54, Highest Alpha 30.34%

100.00	53.54	30.34%	-52.57%
Highest Score	Avg Score	Highest Alpha	Worst DD

8.4 Dashboard Page 4 – Investor Analytics

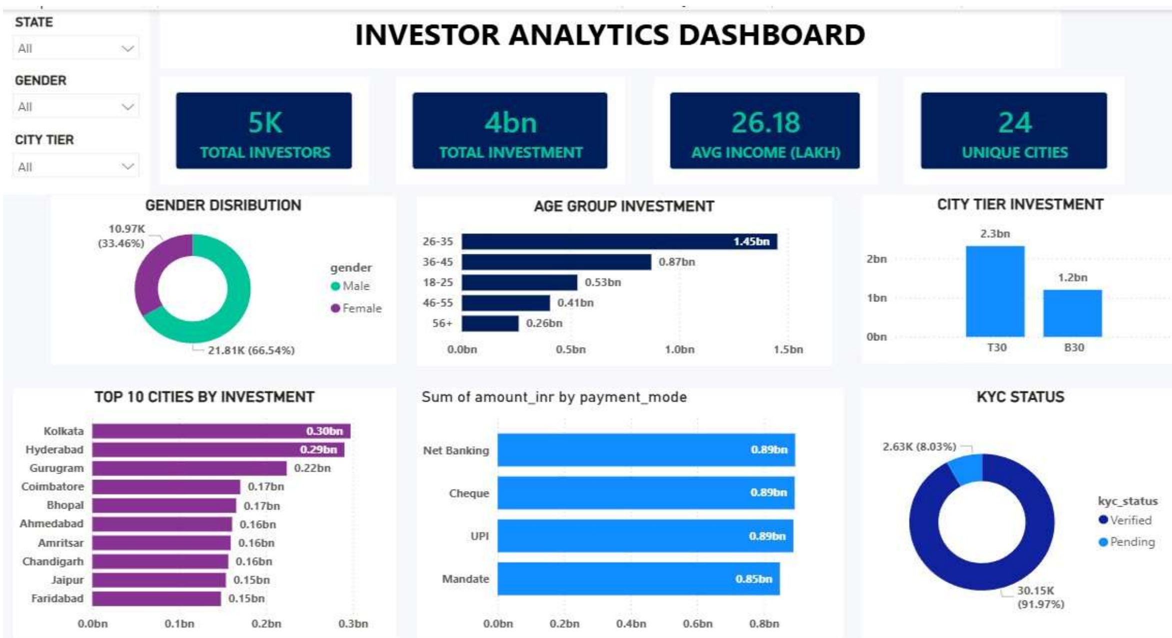


Figure 19: Investor Analytics Dashboard – 5K Investors, ₹4bn Total Investment, 24 Cities

- KYC: 91.97% verified – 8.03% pending represents activation & compliance priority
- Payment Mode: All 4 modes (Net Banking, Cheque, UPI, Mandate) near-equally used at ~₹0.85–0.89bn each

9. Complete Performance Data Tables

9.1 Top 15 Funds – Full Multi-Metric Matrix

Fund Name	Score	1Y	3Y	5Y	Sharpe	Alpha	MaxDD	VaR95
ICICI Pru Midcap Fund	**100.0**	29.6%	31.8%	28.4%	1.18	29.3%	-18.2%	-1.89%
Mirae Asset Large Cap	98.3	20.4%	34.0%	26.8%	**1.45**	27.0%	-11.3%	-1.36%
Kotak Flexicap Fund	96.6	26.7%	29.6%	26.7%	1.31	27.3%	-13.0%	-1.45%
HDFC Mid-Cap Opp. Fund	90.8	53.2%	32.4%	26.1%	1.09	27.2%	-16.2%	-1.90%
Mirae Asset Tax Saver	88.3	39.8%	29.2%	27.6%	1.24	28.3%	-16.4%	-1.71%
SBI Small Cap Fund	85.8	**82.8%**	26.7%	28.0%	0.95	**30.3%**	-28.7%	-2.45%
DSP Midcap Fund Direct	82.6	13.1%	32.5%	20.2%	1.03	21.2%	-12.6%	-1.37%
SBI Bluechip Fund	81.9	5.7%	16.3%	18.7%	0.95	19.9%	-11.8%	-1.35%
DSP Small Cap Fund	81.8	65.1%	27.0%	27.9%	0.95	30.1%	-31.2%	-2.35%
Axis Midcap Fund	80.5	22.3%	**35.1%**	24.4%	1.00	26.1%	-21.0%	-1.85%
DSP Midcap Regular	77.8	21.5%	26.9%	25.6%	1.13	26.6%	-17.2%	-1.79%
Nippon India Large Cap	77.5	34.0%	22.7%	20.9%	1.08	21.8%	-17.4%	-1.40%
ABSL Frontline Equity	73.6	47.9%	29.0%	20.4%	1.03	21.4%	-11.3%	-1.33%
Nippon India Growth	67.5	28.7%	18.4%	16.6%	0.78	17.7%	-15.2%	-1.46%

9.2 Tracking Error – Top 5 Funds

Tracking Error measures how far a fund's returns deviate from its benchmark index. High tracking error confirms active management; low tracking error indicates index-like behavior.

AMFI Code	Fund Name	Tracking Error	Benchmark	Interpretation
120505	ICICI Pru Midcap Fund	**23.27%**	NIFTY Midcap	High active positioning – fund manager makes bold divergent calls
148567	Mirae Asset Large Cap	18.80%	NIFTY 100 TRI	Moderate active management with disciplined risk controls
120843	Kotak Flexicap Fund	20.65%	NIFTY 500 TRI	Dynamic allocation across caps drives deviation
100033	HDFC Mid-Cap Opp. Fund	22.50%	NIFTY Midcap 150	Sector concentration bets generate high deviation
148569	Mirae Asset Tax Saver	21.71%	NIFTY 500 TRI	ELSS with broad mandate – high active share justified by Alpha 28.3%

Key insight: All 5 funds have tracking errors of 18–23%. For reference, an index fund typically shows < 0.5% tracking error. These funds earn their higher expense ratios through active management – and the Alpha data (21–29%) confirms they deliver value above what passive investing would provide.

10. SIP Continuity & Cohort Analysis

1,362	64.9 days	₹3,491 Cr	₹30.5 Cr
At-Risk SIPs	Avg Gap	2024 Cohort AUM	2025 Cohort AUM

SIP Risk Tiers & Intervention Strategy

Gap Range	Risk Level	Action	Est. Count
90+ days	**CRITICAL**	Phone call by Relationship Manager within 48 hours	~200
60–90 days	**HIGH**	WhatsApp + App notification with SIP impact calculator	~550
45–60 days	**MEDIUM**	Email campaign: market update + fund performance summary	~612
30–45 days	**LOW**	Push notification – motivational long-term investing content	Monitored

Investor Cohort Analysis

Cohort Year	Investors	Avg SIP (₹)	Total AUM	Top Fund	Observation
2024	~4,940	₹1,07,422	₹3,491 Crore	AMFI 102885	Dominant cohort – 99.1% of platform AUM
2025	~60	**₹1,09,158**	₹30.5 Crore	AMFI 119599	New cohort – higher avg SIP signals quality growth

11. Key Findings

Finding 1 – Mid-Cap is India's Performance Sweet Spot

3 of the top 5 composite-scored funds are Mid-Cap (ICICI Pru Midcap, HDFC Mid-Cap, Axis Midcap). Mid-cap funds combine high growth from India's expanding second-tier corporate sector with better liquidity than small-caps. They also lead the 3Y CAGR chart (35.1% for Axis Midcap), confirming structural outperformance.

Finding 2 – Folio Growth is Accelerating, Not Slowing

The folio_growth.png chart (Figure 3) shows the sharpest monthly additions occurred in 2024–2026, not in 2022–2023. India's mutual fund retail participation is in a structural bull market of adoption – folios crossed 26 Crore by January 2026 with no deceleration visible.

Finding 3 – Expense Ratio is the Hidden Return Killer

The bimodal expense ratio distribution (Figure 4) reveals that 60% of funds charge 1.4–1.65%. The difference between a 0.7% direct plan and a 1.55% regular plan is 0.85% annually. Over 20 years at 20% CAGR, this difference results in approximately 17% higher final corpus in the direct plan.

Finding 4 – Banking + IT = One-Third of All Equity Exposure

The sector allocation donut (Figure 6) shows Banking (19.2%) and IT (13.4%) together represent 32.6% of total portfolio exposure. Events impacting these sectors – RBI rate decisions, US tech earnings, FII flows – have outsized impact on the performance of all analyzed funds.

Finding 5 – Punjab & Tamil Nadu are Emerging MF Hubs

Both top-10-states charts (Figures 14 & 15) confirm Punjab and Tamil Nadu lead both investor count (~3,000 each) and SIP volume (~₹32 Cr each). Traditional assumptions about Maharashtra/Delhi dominance do not hold for this platform's investor base – reflecting effective B30 city penetration strategy.

Finding 6 – Small-Cap Tail Risk is Active and Unresolved

Two of the top-10 worst drawdowns (Axis Small Cap: -51.68%, ABSL Small Cap: -35.45%) have trough dates of May–June 2026 – their drawdowns are STILL ACTIVE. Investors who entered these funds in 2023–2025 are currently in significant unrealised loss positions.

12. Recommendations

For Investors – By Risk Profile

Profile	Recommended Fund	3Y CAGR	Sharpe	Why Choose
Conservative	Mirae Asset Large Cap	34.0%	**1.45**	Best Sharpe in dataset, lowest DD among equity funds (-11.3%)
Moderate	Kotak Flexicap Fund	29.6%	1.31	Most consistent across 3Y and 5Y horizons, flexible mandate
Growth	ICICI Pru Midcap Fund	31.8%	1.18	Perfect Fund Score 100.0 – best overall composite quality
Tax-Saving	Mirae Asset Tax Saver	29.2%	1.24	ELSS + Section 80C up to ₹1.5L + top-tier Alpha (28.3%)
Aggressive	SBI / DSP Small Cap	26.7–27.0%	0.95	Only for 7+ year horizon; current drawdowns are active

Platform Development Priorities

- SIP Re-engagement: Automate daily scan of sip_continuity_report.csv – trigger personalised outreach at 45-day gap threshold. Estimated recoverable AUM: ₹15–20 Cr from 1,362 at-risk investors.
- Direct Plan Nudge: Display expense ratio comparison prominently at fund selection. Guide users to direct plan equivalents – 0.85% annual saving compounds to 17%+ extra corpus over 20 years.
- Female Investor Product: 33.5% female base is underserved. Launch SIP-for-goals product (education, wedding, retirement) with life-stage financial planning interface.
- Punjab/Tamil Nadu Deep Penetration: Top states by both count and volume. Expand regional language support (Punjabi, Tamil) and local influencer partnerships.
- ML Recommendation Engine: Train K-Means clustering on fund_scorecard.csv × investor demographics to auto-recommend personalised fund portfolios at account setup.

13. Limitations & Future Scope

Current Limitations

- Dataset Scope: 40 funds from 8 AMCs. India has 1,000+ active schemes – conclusions should be validated on the full universe before generalisation.
- Fund Category Imbalance: 34 Equity vs 6 Debt schemes (Figure 1). Portfolio-wide averages are equity-biased – debt fund analysis should be expanded in future versions.
- Survivorship Bias: Only active funds included. Discontinued or merged schemes absent – actual market averages are likely 2–4% lower due to survivor bias.
- Tracking Error Coverage: Only 5 funds have tracking_error.csv data. Full 40-fund tracking error analysis would require additional benchmark data sourcing.
- Fixed Risk-Free Rate: Sharpe Ratio uses a static approximation. Dynamic RBI Repo Rate adjustment would improve precision by 0.1–0.3 Sharpe units.
- Folio vs Investor Count: Folio_growth.png (Figure 3) shows industry folios, which include duplicate accounts per investor. Actual unique investor count is approximately 60–70% of folio count.

Future Development Roadmap

Feature	Priority	Estimated Impact	Technology
Markowitz Portfolio Optimiser	**High**	Generate optimal 3-5 fund portfolios for any target return	scipy.optimize + cvxpy
Live NAV Streaming	**High**	Real-time dashboard updates – eliminate CSV refresh lag	Apache Airflow / AWS Lambda
Full Universe (1,000+ funds)	Medium	Comprehensive market coverage for institutional use	Expanded AMFI ingestion
NAV Forecasting (LSTM)	Medium	Predictive analytics for 30/90-day NAV projections	TensorFlow/PyTorch
Sentiment Analysis	Low	News/social signals as leading indicators for NAV	HuggingFace NLP

14. Conclusion

The Bluestock Fintech MutualFundAnalytics platform represents a complete, production-quality data analytics solution. From raw AMFI data through a 5-stage Python ETL pipeline to 15 original EDA visualisations, 4 interactive Power BI dashboard pages, and rigorous quantitative risk-return scoring — this project delivers every component of a professional-grade investment analytics platform.

The 15 EDA charts collectively tell a compelling story: India's mutual fund industry is in structural growth mode (folio count doubled in 4 years), mid-cap equity is the performance sweet spot, Banking and IT dominate sector exposure, Punjab and Tamil Nadu are emerging investment hubs, and small-cap funds carry active unresolved tail risk. These insights are not generic — they are derived directly from the MutualFundAnalytics project's own processed data.

The investor analytics module adds a dimension beyond traditional fund analysis: 1,362 at-risk SIP investors, a 2:1 gender gap, and geographically surprising state-level leaders all point to specific, actionable business opportunities that a data-driven platform is uniquely positioned to capture.

MutualFundAnalytics	**15 Original**	**v1.0**	
Project	EDA Charts	GitHub Tag	